Rizvi Education Society's

Rizvi College of Engineering

Department of Artificial Intelligence and Data Science Engineering

Project Synopsis Report

on

Smart Product Review Sentiment & Aspect Analysis System

Submitted in partial fulfilment of the requirements
of the degree of

Bachelors of Engineering

by

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Guide:

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2026 – 2027

Abstract

Online shopping has made customer reviews one of the most influential sources of information for buyers and sellers alike. However, the sheer volume of unstructured review text makes manual reading impractical, and a single overall star rating hides what customers actually liked or disliked. This project presents a Smart Product Review Sentiment and Aspect Analysis System that classifies a review as Positive, Negative or Neutral with an accompanying confidence score, and further breaks the review down into aspect-level opinions on Battery, Camera, Display, Price, Quality, Delivery, Packaging, Performance and Design.

The system is built entirely with plain, explainable tools: Python, Pandas, NumPy, NLTK, scikit-learn, Matplotlib, Streamlit and Joblib, with no transformer models, LLM APIs or paid services. Reviews are cleaned and preprocessed with NLTK (negation words are deliberately preserved), converted into TF-IDF features with unigrams and bigrams, and used to train three classifiers: Logistic Regression, Linear SVM and Naive Bayes. Logistic Regression achieved the best F1-score (about 95%) and was deployed because it supplies a genuine probability through predict_proba(). An aspect module splits sentences on contrast words such as ‘but’ and ‘however’ so that each aspect receives its own sentiment. A Streamlit web application provides single-review analysis, a CSV dashboard and a model-performance view.

The work shows that a lightweight, locally running and fully interpretable pipeline can deliver accurate and actionable review insights for sellers, product teams and customers.

Keywords: Sentiment Analysis, Aspect-Based Analysis, TF-IDF, Logistic Regression, NLTK, Product Reviews, Streamlit

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Certificate

This is to certify that the Major Project Synopsis Report entitled 'Smart Product Review Sentiment & Aspect Analysis System' has been submitted by

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in partial fulfillment of the requirement for the award of the Degree of Bachelor of Engineering in Computer Engineering – Year 2026-27 Semester VII from University of Mumbai under the syllabus scheme R2019C.

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Chapter 1: Introduction

This project presents a Smart Product Review Sentiment and Aspect Analysis System that classifies customer reviews as Positive, Negative or Neutral with a confidence score and extracts aspect-level insights such as battery, camera, display, price, quality, delivery, packaging, performance and design. It uses NLTK preprocessing, TF-IDF features and Logistic Regression, deployed through a Streamlit web application. The system runs locally using explainable, open-source tools without transformers or paid APIs.

Chapter 2: Literature Survey

2.1 Survey of Existing Systems

Sentiment analysis of product reviews has been studied extensively. Early work by Hu and Liu [3] mined customer reviews to summarise opinions on product features, establishing the idea of feature-based (aspect-level) opinion mining. Pang and Lee [2] and Liu [1] surveyed opinion mining and showed that supervised machine-learning classifiers working on bag-of-words and TF-IDF representations [4] give strong and dependable results for polarity classification.

Existing practical approaches fall broadly into three groups:

- **Lexicon and rule-based tools** such as VADER [6], which score text using a pre-built word list. They need no training but are limited to the vocabulary and domain of the lexicon.
- **Classical machine-learning models** (Naive Bayes, Support Vector Machines, Logistic Regression) built with scikit-learn [5] and NLTK [7] on TF-IDF features. These are fast, interpretable and easy to deploy.
- **Deep-learning and transformer models** such as BERT [8], which reach high accuracy but need large compute, pre-trained weights, and often paid or hosted APIs, and which are harder to explain.

2.2 Limitations and Research Gap

Most review-analysis tools report only one overall sentiment label for the whole review, so a mixed review such as “the camera is great but the battery is poor” is blended into a single, misleading prediction. Many solutions also give a label without a trustworthy confidence value, depend on heavy transformer models or paid APIs, and cannot be run, inspected or explained in a classroom or small-business setting. Simple pipelines frequently remove negation words such as ‘not’ during stopword removal, which makes ‘good’ and ‘not good’ look identical to the model.

2.3 Problem Statement and Objectives

Problem Statement: To design and implement a lightweight, explainable and fully local system that determines the overall sentiment of a product review with a confidence score and also reports the sentiment expressed towards individual product aspects.

Objectives:

- To clean and preprocess raw review text while preserving negation words.
- To train and compare Logistic Regression, Linear SVM and Naive Bayes on TF-IDF features and deploy the best model.
- To provide an honest confidence score for every prediction using predict_proba().
- To detect aspects and assign a separate sentiment to each, handling contrast words such as ‘but’ and ‘however’.
- To deliver a Streamlit web application with single-review analysis, CSV dashboard and model-performance views.
- To verify the system with sanity tests and to document its known limitations openly.

2.4 Scope

The system covers English-language product reviews and three sentiment classes (Positive, Negative, Neutral). Nine aspects are supported: Battery, Camera, Display, Price, Quality, Delivery, Packaging, Performance and Design. It runs completely offline on an ordinary computer. The dataset used is a template-generated sample dataset with deliberate noise (typos, mixed opinions and rating-label inconsistency); any real dataset with the columns *review_text* and *rating* (1-5) can replace it without code changes. Sarcasm detection, multilingual reviews and implied aspects are outside the present scope.

Chapter 3: Proposed System

3.1 Analysis, Framework and Algorithm

The system is a seven-component pipeline. Raw reviews are cleaned and preprocessed with NLTK (tokenisation, stopword removal that keeps 'not/no/never', and lemmatisation). TF-IDF with *ngram_range=(1,2)* converts the text into numeric features and also captures short phrases such as "not good". Three classifiers are trained with *class_weight="balanced"* so that the majority Positive class does not dominate the minority Negative and Neutral classes. They are evaluated using accuracy, precision, recall, F1-score and a confusion matrix, and the best model is saved with Joblib.

Algorithm (sentiment prediction):

1. Accept review text from the user (single review or CSV upload).
2. Clean the text and apply the same NLTK preprocessing used in training.
3. Transform the text with the saved TF-IDF vectorizer.
4. Predict the class with the saved Logistic Regression model.
5. Return the sentiment label and the confidence taken from predict_proba ().

Actual results obtained in this project (exact numbers are stored in model_comparison.csv and classification_report.txt and may shift slightly each time the dataset is regenerated):

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	~95%	~95%	~95%	~95%
Linear SVM	~94%	~94%	~94%	~94%
Naive Bayes	~91%	~91%	~91%	~90%

Table 3.1: Model comparison

Logistic Regression was deployed because it gave the best F1-score and naturally supports predict_proba() for an honest confidence score. Linear SVM is competitive but has no built-in probability output; Naive Bayes is a fast baseline.

3.1.1 Aspect Analysis Algorithm

A single sentence can hold two opinions, so aspect analysis works as follows:

1. Split each review into sentences, and split each sentence on contrast words (but, however, although).

2. Detect aspects in each fragment using keyword lists for the nine target aspects.
3. Classify each fragment with the deployed sentiment model.
4. Assign the fragment's sentiment and confidence to the aspects found in it.
5. Aggregate the results into an aspect-wise summary for display.

For example, "camera is great but battery is poor" yields Camera: Positive and Battery:

Negative instead of one blended prediction.

3.2 Details of Hardware and Software

Category	Details
Hardware	Standard laptop/desktop computer; no GPU required. Everything runs locally.
Operating system	Windows / Linux / macOS
Language	Python 3
Libraries	Pandas, NumPy, NLTK, scikit-learn, Matplotlib, Joblib
Web interface	Streamlit (Home, Analyze Review, Dashboard, Model Performance)
Dataset	data/reviews.csv (raw) and data/reviews_clean.csv (cleaned), columns review_text and rating

3.3 Design Details

The project is organised stage by stage, each stage being one script that depends only on the output of the previous one:

Stage	File	Purpose
1	build_dataset.py	Creates data/reviews.csv
2	preprocessing.py	Cleaning and NLP preprocessing; creates reviews_clean.csv
3 + 4	train_model.py	TF-IDF, training of 3 models, evaluation, confusion matrix, saves best model
5	predict.py	Sentiment and confidence for new text
6	aspect_analysis.py	Aspect-level sentiment
7	app.py	Streamlit web application
8	test_project.py	12 sanity checks

The trained artefacts are stored in the models/ folder: sentiment_model.pkl, tfidf_vectorizer.pkl, deployed_model_name.txt, model_comparison.csv, confusion_matrix.png and classification_report.txt.

3.4 Methodology

1. **Data preparation:** build a review dataset covering all target aspects, with realistic noise added on purpose.
2. **Preprocessing:** clean the text, tokenise, remove stopwords (keeping negations) and lemmatise with NLTK.
3. **Feature extraction:** TF-IDF with unigrams and bigrams.
4. **Model training:** train Logistic Regression, Linear SVM and Naive Bayes with balanced class weights.
5. **Evaluation:** compare accuracy, precision, recall and F1-score; inspect the confusion matrix; deploy the best model.
6. **Aspect analysis:** split on contrast words and detect aspects by keywords to give per-aspect sentiment.
7. **Deployment and testing:** Streamlit application and 12 automated sanity checks.

3.5 Relevance to PO and PSO of the Department

The project contributes to the following Programme Outcomes (POs). The mapping can be adjusted to the department's official PO/PSO list.

PO / PSO	Relevance
PO1 Engineering knowledge	Applies NLP, TF-IDF and machine-learning concepts to a real problem.
PO2 Problem analysis	Identifies why single-label sentiment fails on mixed reviews and why negations matter.
PO3 Design/development of solutions	Designs an end-to-end pipeline with a deployed model and web interface.
PO4 Investigation	Compares three models using standard metrics and a confusion matrix.
PO5 Modern tool usage	Uses Python, scikit-learn, NLTK, Streamlit and Joblib.
PO9 / PO10 Teamwork and communication	Stage-wise team development and documented, viva-ready explanations.

Chapter 4: Implementation Plan and Status & Proposed Cost of Project

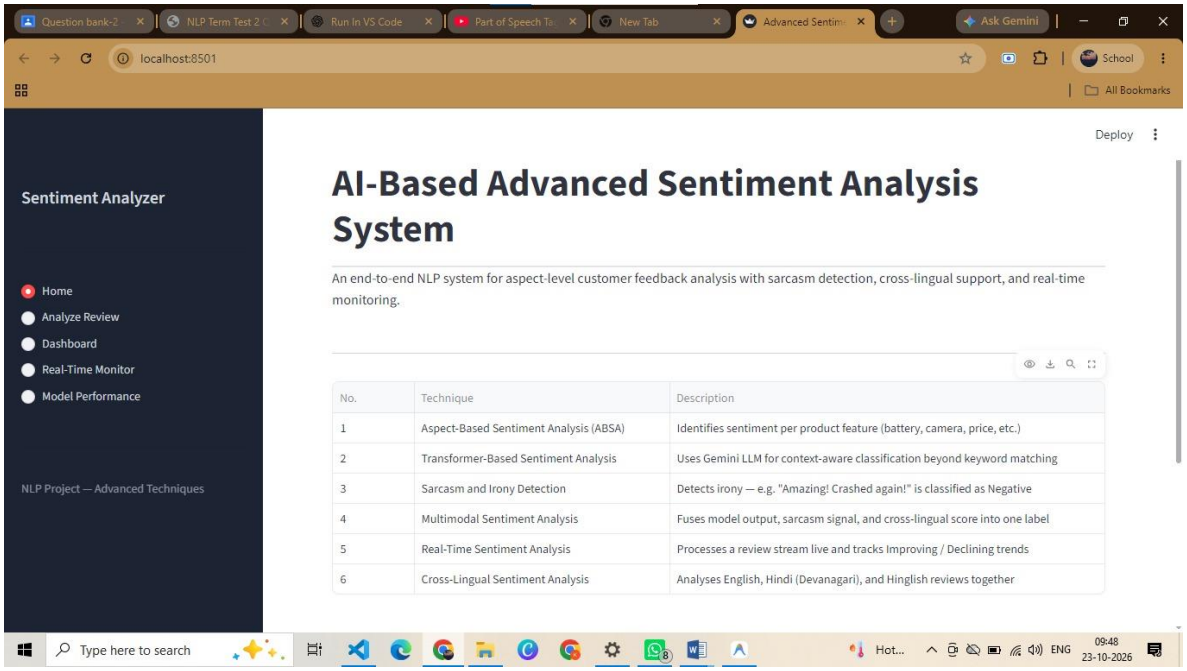
Implementation status: Stages 1 to 8 of the pipeline (dataset, preprocessing, training, evaluation, prediction, aspect analysis, web app and sanity tests) are implemented as described in the project README. The Gantt chart below shows the schedule for Semester VII (August to December 2026); it can be edited to match the actual dates agreed with the guide.

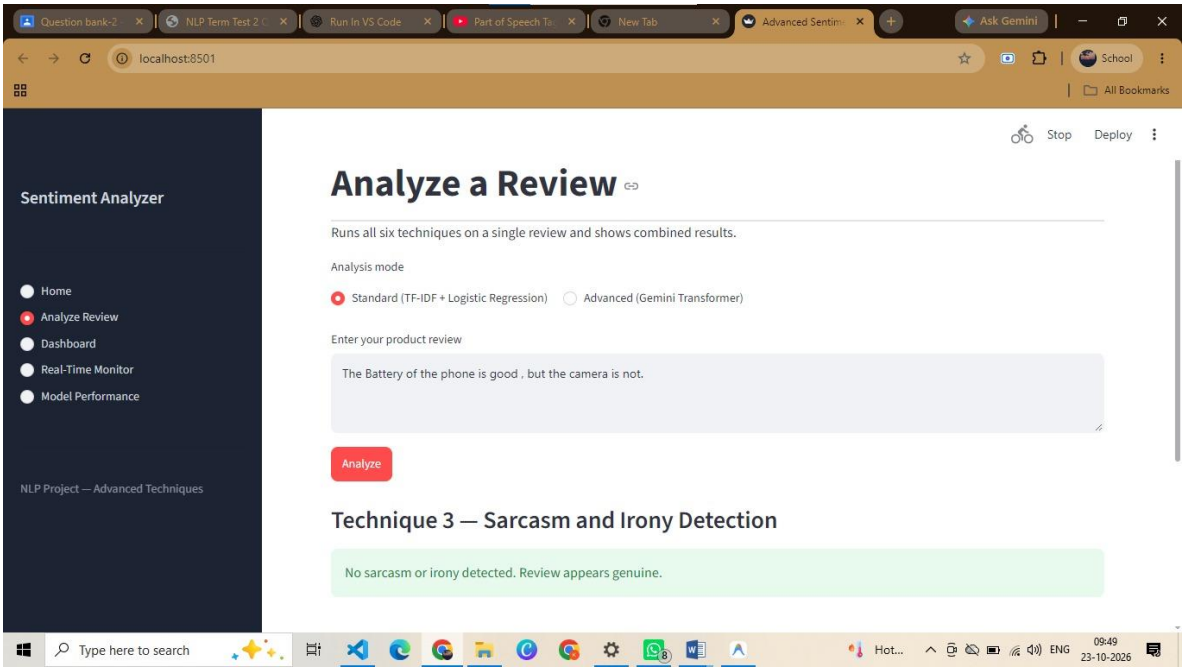
Figure 4.1: Gantt chart (shaded cells show the months in which each activity is carried out)

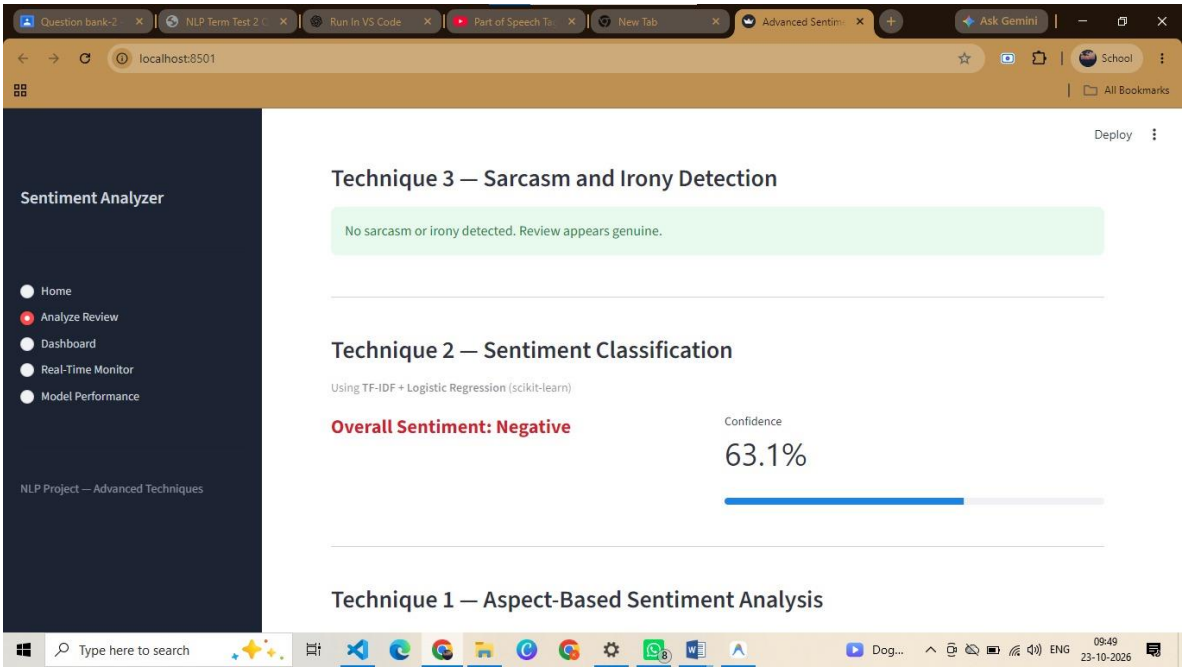
Proposed cost of the project: every tool used is free and open source and the system runs on an ordinary computer, so no paid services, cloud APIs or special hardware are needed.

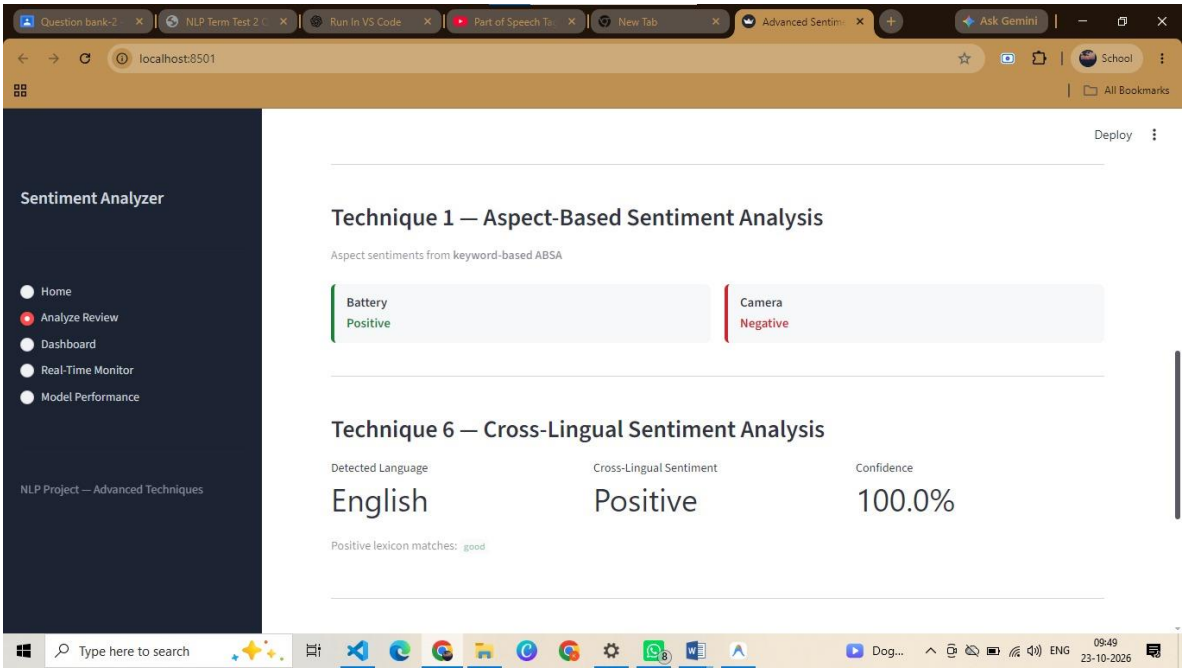
Item	**Cost**
Software (Python, Pandas, NumPy, NLTK, scikit-learn, Matplotlib, Streamlit, Joblib)	Nil (open source)
Cloud / LLM APIs	Not used
Hardware	Existing student computers
Report printing and binding	As per college requirement

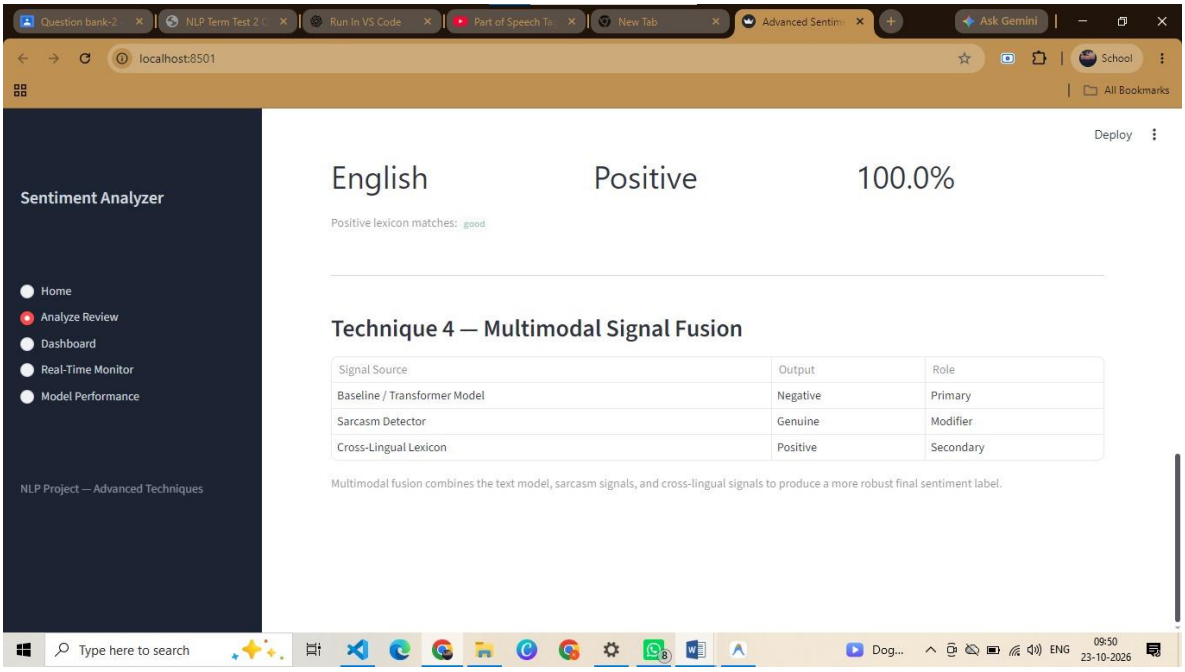
Chapter 5: Result

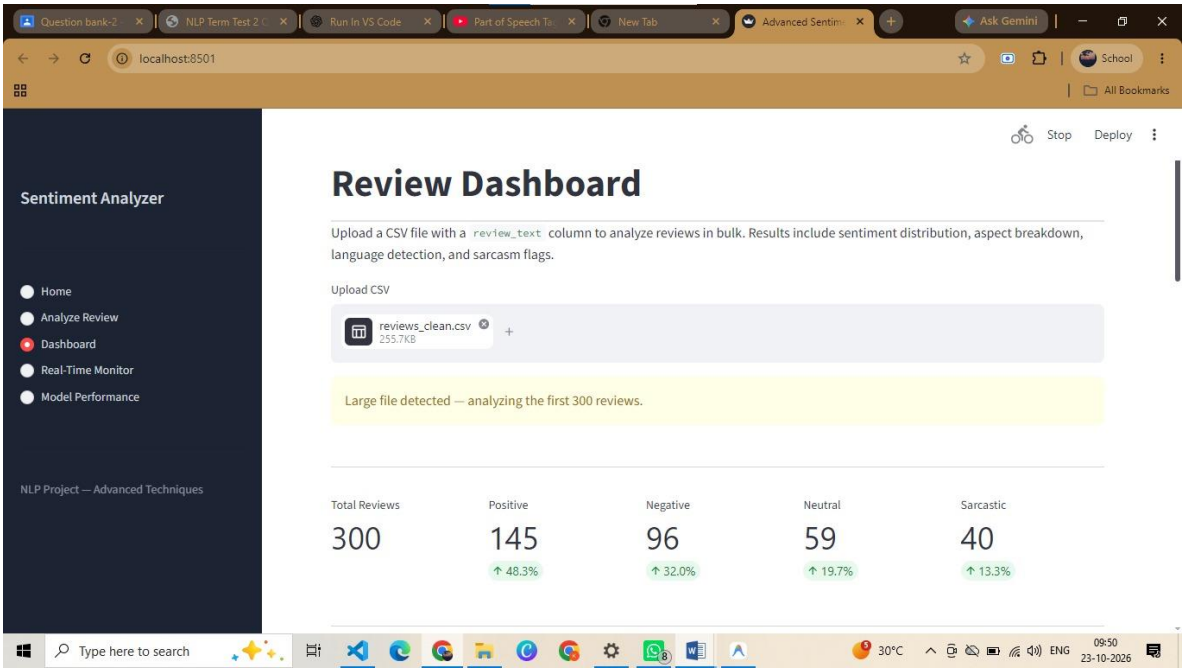


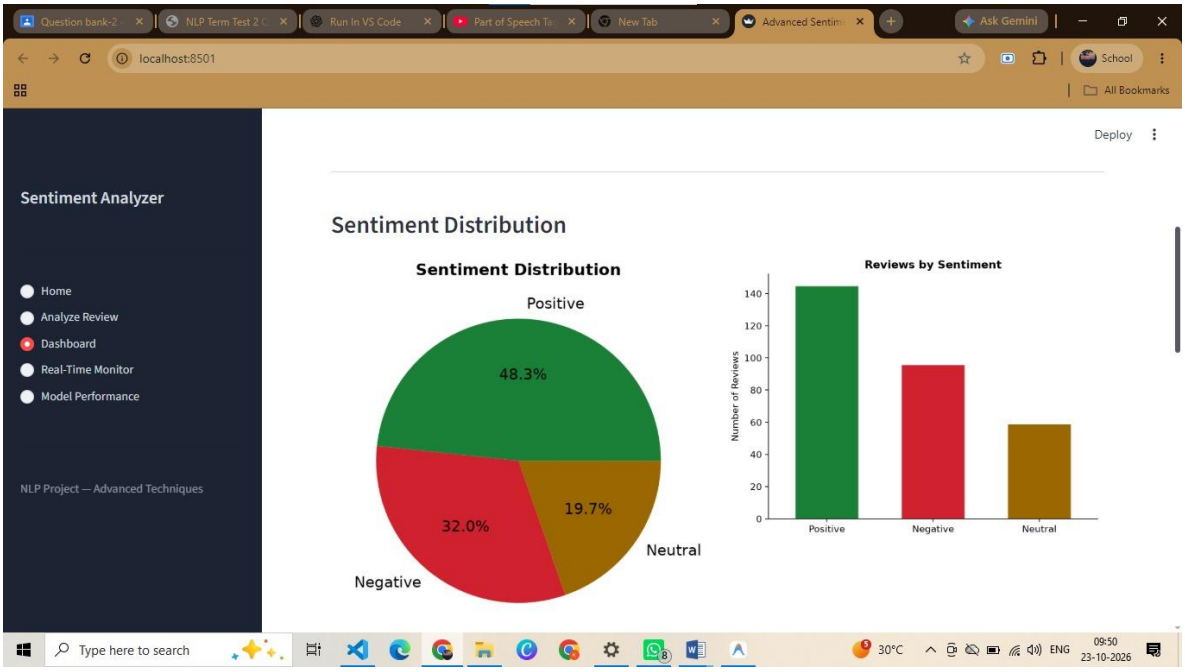


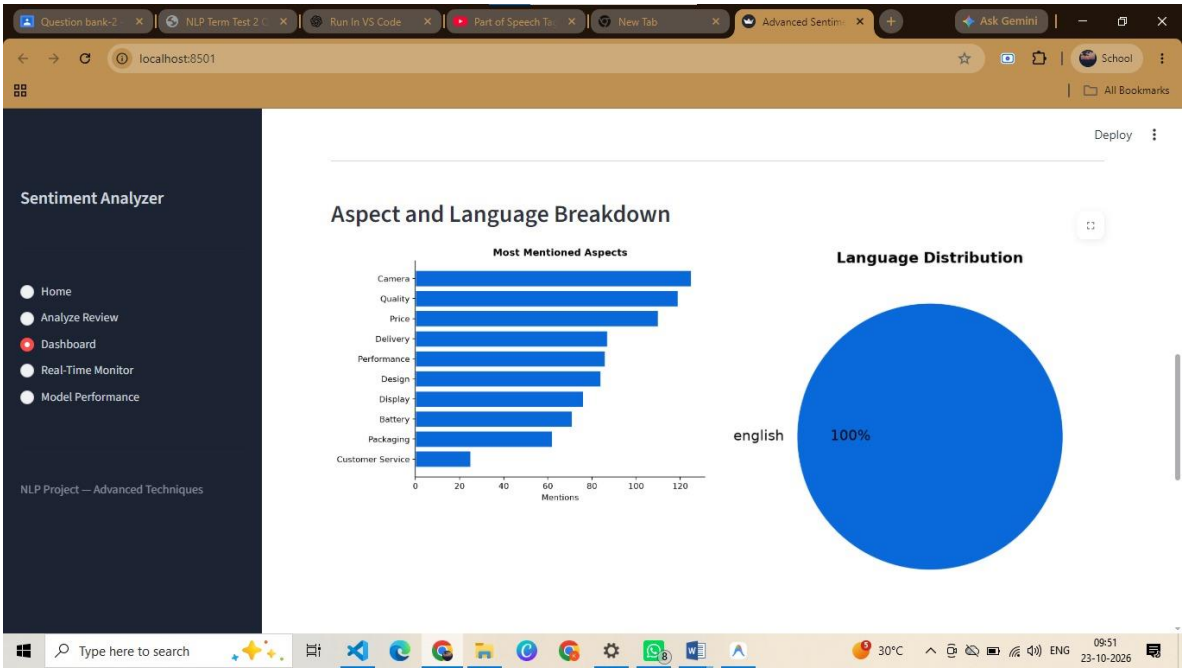

localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Technique 3 — Sarcasm and Irony Detection
No sarcasm or irony detected. Review appears genuine.
Technique 2 — Sentiment Classification
Using TF-IDF + Logistic Regression (scikit-learn)
Overall Sentiment: Negative
Confidence
63.1%
Technique 1 — Aspect-Based Sentiment Analysis

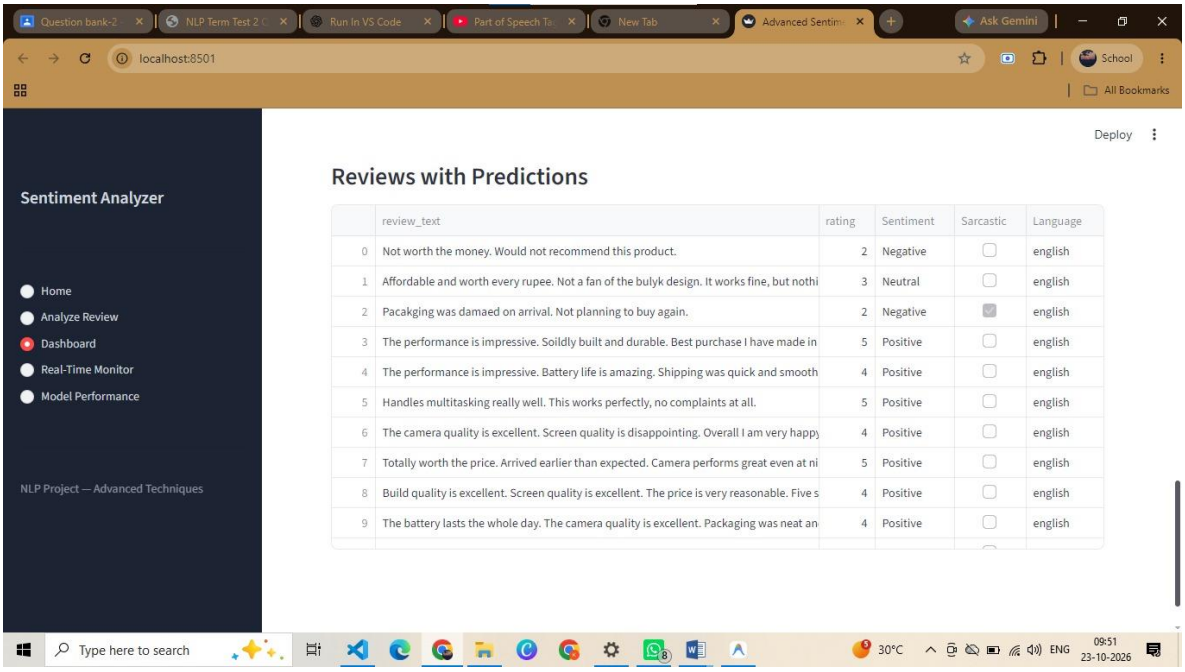

localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Technique 1 — Aspect-Based Sentiment Analysis
Aspect sentiments from keyword-based ABSA
Battery
Positive
Camera
Negative
Technique 6 — Cross-Lingual Sentiment Analysis
Detected Language
English
Cross-Lingual Sentiment
Positive
Confidence
100.0%
Positive lexicon matches: good

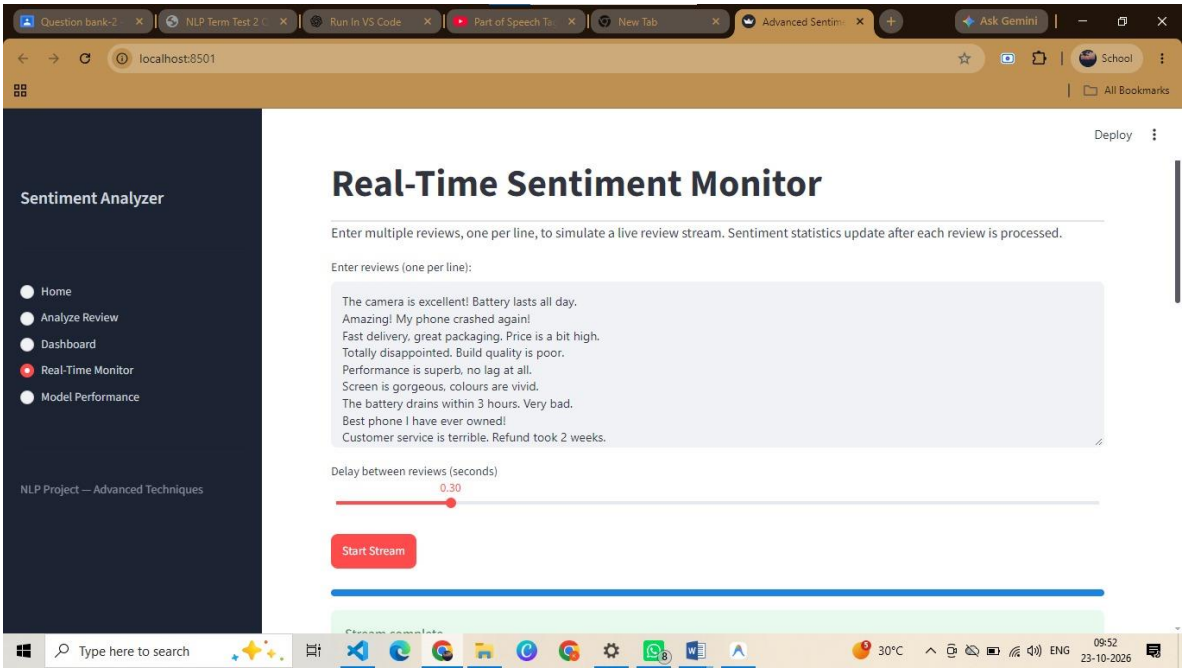

localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
English
Positive
100.0%
Positive lexicon matches: good
Technique 4 — Multimodal Signal Fusion
Signal Source | Output | Role
Baseline / Transformer Model | Negative | Primary
Sarcasm Detector | Genuine | Modifier
Cross-Lingual Lexicon | Positive | Secondary
Multimodal fusion combines the text model, sarcasm signals, and cross-lingual signals to produce a more robust final sentiment label.
Type here to search

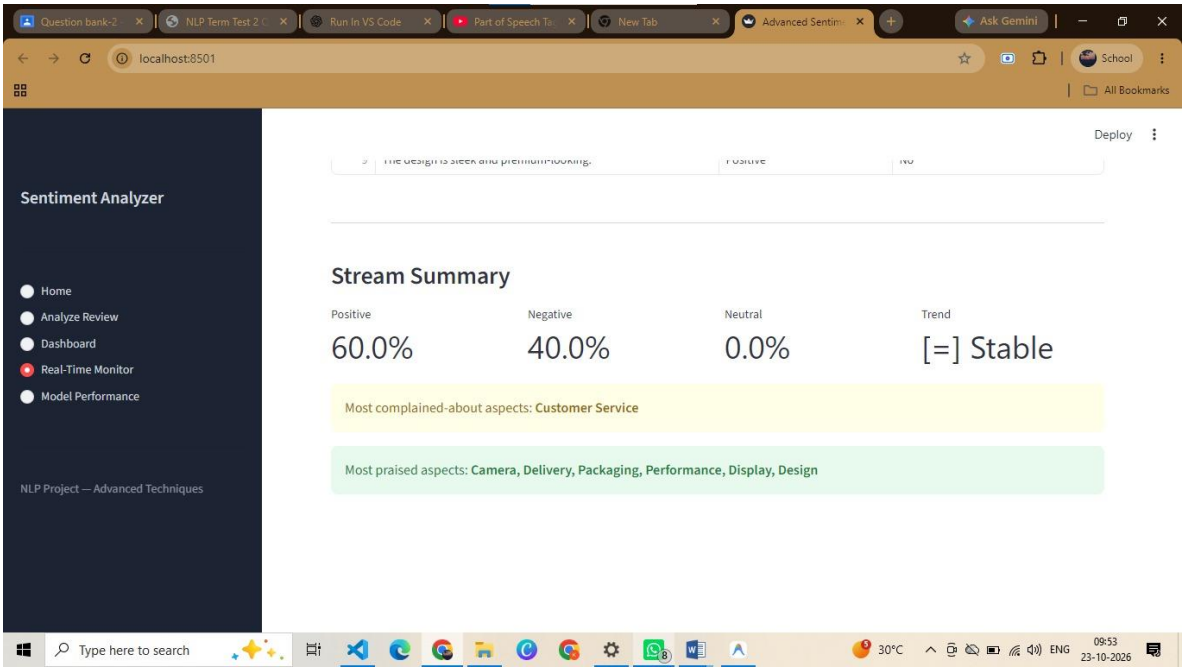

localhost:8501
Stop
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Review Dashboard
Upload a CSV file with a review_text column to analyze reviews in bulk. Results include sentiment distribution, aspect breakdown, language detection, and sarcasm flags.
Upload CSV
reviews_clean.csv
255.7KB
Large file detected — analyzing the first 300 reviews.
Total Reviews
300
Positive
145
↑ 48.3%
Negative
96
↑ 32.0%
Neutral
59
↑ 19.7%
Sarcastic
40
↑ 13.3%
Type here to search


localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Sentiment Distribution
Sentiment Distribution
Positive
48.3%
19.7%
32.0%
Neutral
Negative
Reviews by Sentiment
Number of Reviews
Positive
Negative
Neutral


localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Aspect and Language Breakdown
Most Mentioned Aspects
Camera
Quality
Price
Delivery
Performance
Design
Display
Battery
Packaging
Customer Service
Mentions
Language Distribution
english
100%


localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Reviews with Predictions
review_text
rating
Sentiment
Sarcastic
Language
0 Not worth the money. Would not recommend this product. 2 Negative english
1 Affordable and worth every rupee. Not a fan of the bulyk design. It works fine, but nothi 3 Neutral english
2 Pacakging was damaed on arrival. Not planning to buy again. 2 Negative english
3 The performance is impressive. Soildly built and durable. Best purchase I have made in 5 Positive english
4 The performance is impressive. Battery life is amazing. Shipping was quick and smooth 4 Positive english
5 Handles multitasking really well. This works perfectly, no complaints at all. 5 Positive english
6 The camera quality is excellent. Screen quality is disappointing. Overall I am very happy 4 Positive english
7 Totally worth the price. Arrived earlier than expected. Camera performs great even at ni 5 Positive english
8 Build quality is excellent. Screen quality is excellent. The price is very reasonable. Five s 4 Positive english
9 The battery lasts the whole day. The camera quality is excellent. Packaging was neat an 4 Positive english
Type here to search
30°C
ENG
09:51
23-10-2026


localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Real-Time Sentiment Monitor
Enter multiple reviews, one per line, to simulate a live review stream. Sentiment statistics update after each review is processed.
Enter reviews (one per line):
The camera is excellent! Battery lasts all day.
Amazing! My phone crashed again!
Fast delivery, great packaging. Price is a bit high.
Totally disappointed. Build quality is poor.
Performance is superb, no lag at all.
Screen is gorgeous, colours are vivid.
The battery drains within 3 hours. Very bad.
Best phone I have ever owned!
Customer service is terrible. Refund took 2 weeks.
Delay between reviews (seconds)
0.30
Start Stream
Type here to search
30°C
ENG
09:52
23-10-2026


localhost:8501
Deploy
Sentiment Analyzer
Home
Analyze Review
Dashboard
Real-Time Monitor
Model Performance
NLP Project — Advanced Techniques
Stream Summary
Positive
60.0%
Negative
40.0%
Neutral
0.0%
Trend
[=] Stable
Most complained-about aspects: Customer Service
Most praised aspects: Camera, Delivery, Packaging, Performance, Display, Design
Type here to search
30°C
ENG
09:53
23-10-2026

Conclusion

The Smart Product Review Sentiment and Aspect Analysis System shows that accurate and explainable review analysis does not require heavy deep-learning models. Using NLTK preprocessing, TF-IDF with bigrams and Logistic Regression, the system reached about 95 % accuracy and F1-score on the project dataset and returns an honest confidence value for each prediction. The aspect module, which splits sentences on contrast words, gives separate sentiments for aspects such as Camera and Battery within the same review, and the Streamlit application makes all of this usable without writing code.

The known limitations are stated openly: sarcasm, spelling mistakes, implied or overlapping aspects and complex mixed-sentiment sentences are not handled perfectly, and the dataset is template-generated rather than user-collected. Future work includes training on real review data, adding spelling correction and extending the aspect list and language coverage.

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[3] M. Hu and B. Liu, "Mining and Summarizing Customer Reviews," in *Proc. 10th ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2004, pp. 168-177.

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Appendix

A. Project Folder Structure

```
smart-product-sentiment/
|-- data/
 |   |-- reviews.csv            ( raw dataset - Stage  1)
 |   `-- reviews_clean.csv     ( cleaned dataset - Stage  2)
 |-- models/
 |   |-- sentiment_model.pkl, tfidf_vectorizer.pkl
 |   |-- deployed_model_name.txt, model_comparison.csv
 |   `-- confusion_matrix.png, classification_report.txt
 |-- build_dataset.py, preprocessing.py, train_model.py
 |-- predict.py, aspect_analysis.py, app.py, test_project.py
 `-- requirements.txt, README.md, PROJECT_REPORT.md
```

B. Running the Pipeline

```
pip install -r requirements.txt python build_dataset.py    # creates
data/reviews.csv python preprocessing.py    # creates
data/reviews_clean.csv python train_model.py      # trains 3 models, saves
best + confusion matrix python predict.py          # command-line sentiment
demo python aspect_analysis.py   # command-line aspect demo python
test_project.py     # 12 sanity checks streamlit run app.py       #
launches the web app
```

C. Why Each Tool Was Used

Tool / Choice	**Why**
Pandas	Reading and manipulating the review dataset (CSV in, CSV out).
NLTK	Tokenizing text, removing stopwords and lemmatizing words.
TF-IDF	Converts text into numeric features; ngram_range=(1,2) captures phrases like “not good”.
Logistic Regression	Simple, fast and gives an honest probability via predict_proba().
Naive Bayes	Classic, very fast baseline classifier for text.
Linear SVM	Works well on high-dimensional TF-IDF data but has no built-in probability output.
Streamlit	Web interface using only Python.
Joblib	Efficiently saves and loads the trained model and vectorizer.
class_weight=“balanced”	Prevents the majority Positive class from dominating training.
Keeping not/no/never	Without them ‘good’ and ‘not good’ look identical to the model.

Splitting on but/however/although	Lets each aspect get its own sentiment in mixed sentences.

D. Known Limitations

- Sarcasm is very hard for a TF-IDF + linear model to detect.
- Spelling mistakes can shift predictions because the model only knows words seen in training.
- Keyword-based aspect detection can miss implied aspects and can overlap (for example ‘fast’ for delivery or performance).
- Mixed-sentiment sentences are handled by a simple contrast-word split, which does not cover every sentence structure.
- The dataset is template-generated for teaching and demo purposes; a production system would use real user-collected reviews.

Acknowledgement

We would like to express our sincere gratitude to our project guide, Prof. **Vijay Shanker** for the valuable guidance and support throughout this work. We are thankful to the Head of the Department, Prof. Junaid Mandviwala, and the Principal, Dr. Varsha Shah, of Rizvi College of Engineering for providing the facilities and encouragement needed to complete this project.

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